

Technical supplement

Allometric calibration bias in tropical forest carbon

Derivations, material-accounting sensitivities and source-table provenance

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This supplement supplies the detailed calculations supporting the article's measurement-to-accounting argument. Conditional material requirements, source-table corrections and synthetic tests are distinguished from field observations. The published pantropical total is preserved; no new class-5 field inventory is asserted. Unprefixed references to article equations use the accompanying compiled article; supplement equations and tables carry an S prefix.

S1 Material-accounting definitions and conditional class-density comparisons

The protocol has four linked operations: fix the spatial and carbon-pool target; independently measure geometry and material properties; reconstruct carbon using disjoint compartments and representative plot weights; and compare the resulting class mean with the mapped target under joint uncertainty. Its empirical test is the comparison between independently obtained estimates. Re-expressing one set of measurements in equivalent algebraic forms provides internal consistency, rather than independent validation. The carbon assigned to a mapped class can be translated into material-volume requirements. Class 5 in the source table contains 285.03 PgC over 531,449,420.52 ha, implying 536.3257 MgC ha⁻¹ [1, Table 3]. Its stock is numerically 99.975% of Feldpausch et al.'s entire 285.1 PgC total, on an area 31.871% as large as that comparison's 1667.5 million hectares [2]. The area is 5.31 million km², comparable in scale to the approximately 5.3 million km² of Amazonian tropical forest described by Fauset et al. [3]; this is a historical scale comparison, not a claim of matching masks or current forest area. This spatial concentration is a target for structural measurement, rather than an identified local correction factor between different masks. The combined classes 4 and 5 have a different mean, 472.19 MgC ha⁻¹, and must be distinguished from class 5 alone.

For a sampled plot of area A_p in hectares, let $a_j > 0$ be a consistently measured stem basal cross-section in m², $H_j > 0$ its height in metres, and $V_j \geq 0$ the reconstructed solid volume of its assigned woody compartments in m³. Stem and branch compartments must be disjoint; cavities and external envelope space are excluded from solid material. Crown-boundary and multistem assignment conventions must be fixed. Define

$$\psi_j = \frac{V_j}{a_j H_j}, \quad G_{b,p} = \frac{\sum_j a_j}{A_p} = \nu_p \bar{a}_p, \quad \Lambda_p = \frac{\sum_j a_j H_j \psi_j}{\sum_j a_j}, \quad (\text{S1})$$

where $\nu_p = n_p/A_p$ is stems per hectare. Λ_p is a basal-area-weighted height–architecture quantity in metres. The dimensionless ψ_j is calculated from independent geometry; it is not a fitted biomass allometry or necessarily a stem-only form factor bounded by one. Branch volume can change its interpretation. Different stem-size distributions may yield the same basal area, so the inventory must retain individual or size-class measurements as well as the aggregate.

The height and architectural contributions separate exactly as

$$\bar{H}_a = \frac{\sum_j a_j H_j}{\sum_j a_j}, \quad \bar{\psi}_{aH} = \frac{\sum_j a_j H_j \psi_j}{\sum_j a_j H_j}, \quad \Lambda = \bar{H}_a \bar{\psi}_{aH}.$$

The first mean is basal-area weighted; the second uses basal-area–height weights. An unweighted mean height cannot be substituted without additional assumptions. At $\bar{H}_a = 30$ m, $G_b = 30$ or $40 \text{ m}^2 \text{ ha}^{-1}$ would require $\bar{\psi}_{aH} \approx 2.1$ or 1.6 for the ratio-based class-5 target under the illustrative zero-remainder properties. These are ratios of reconstructed solid woody volume to the summed basal-area–height cylinders, not canopy-envelope occupancy or observed forest parameters. Stem taper, branch volume, buttresses, height and cavity treatment must therefore be resolved independently.

Let $\beta_p = \sum_{j,o \in W} V_{jo} \rho_{jo}^b f_{jo}^C / \sum_{j,o \in W} V_{jo}$ be the measured, volume-weighted carbon density of woody material in MgC m^{-3} . Let $c_{\text{rem},p}$ contain the carbon per hectare in all explicitly excluded aboveground compartments, such as separately measured foliage. Bark belongs either in W with matched properties or in the remainder, never in both. Assume total woody volume is positive.

Measurement definitions and carbon reconstruction. For these matched measurements,

$$\frac{V_{W,p}}{A_p} = G_{b,p} \Lambda_p, \quad \boxed{\bar{c}_p = \beta_p G_{b,p} \Lambda_p + c_{\text{rem},p}} \quad (\text{S2})$$

The volume equality follows directly from the definition of ψ_j ; the carbon equality is the sum of material carbon in disjoint compartments. The procedure is therefore tested against independent observations and the mapped prediction, rather than by treating this internal identity as a new theorem. Geometry and material properties must be measured without fitting ψ or β to the target. Held-out plots, fixed compartment rules and shared measurement uncertainty are part of the test.

Table S1: Conditional physical requirements from the published class-5 lower interval edge, both mean-density conventions and the combined classes. The calculation assumes uniform $f_C = 0.47$, basic density $\rho^b = 0.60 \text{ Mg m}^{-3}$ and zero remainder. Biomass is in Mg ha^{-1} , volume in $\text{m}^3 \text{ ha}^{-1}$ and equivalent-layer depth in cm. Material properties are illustrative, not measured class means. Approximate biomass and volume are rounded to tens and layer depth to one decimal; full calculations are archived. The lower edge constrains mapped-class cells only if the published interval implements the raster classification; it is not a measured lower bound on every plot or subpixel patch.

Domain	MgC ha^{-1}	Dry biomass	Woody volume	Layer depth
Class 5: lower class edge	478	~1,020	~1,700	~17.0
Class 5: printed mean	531.05	~1,130	~1,880	~18.8
Class 5: stock / area	536.33	~1,140	~1,900	~19.0
Classes 4 and 5	472.19	~1,000	~1,670	~16.7

Under the illustrative properties in Table S1, $\beta = 0.282 \text{ MgC m}^{-3}$. The two class-5 mean conventions require approximately 1,880 and $1,900 \text{ m}^3 \text{ ha}^{-1}$, respectively; the approximately one-percent normalization difference leaves the material scale nearly unchanged. The published class interval is 478–605 MgC ha^{-1} [1, Table 3]. At its lower edge, the same properties imply about 1,020 Mg dry biomass and $1,700 \text{ m}^3$ of woody material per hectare. If this interval determines raster membership, those lower-edge requirements apply to the *mapped cell estimates* in class 5, not merely their mean. A mapped cell may contain lower-density patches, and an operational map threshold is not an independently observed biological minimum. Verification of the implemented threshold requires the classified raster and area convention.

For spatial strata s within class 5, with independently specified area weights $\omega_s \geq 0$ summing to one, a class-level reconstruction is

$$\widehat{c}_{5,\text{geom}} = \sum_s \omega_s \overline{\beta_p G_{b,p} \Lambda_p + c_{\text{rem},p}}_s, \quad R_5 = \widehat{c}_{5,\text{geom}} - c_{5,\text{map}}. \quad (\text{S3})$$

The bar denotes the design-weighted mean of plot-level products within a stratum. Multiplying separate means of β , G_b and Λ loses their dependence and generally gives a different answer. The published class ratio is used for $c_{5,\text{map}}$ only on matched class, epoch, pool and area definitions. A representative class mean is the target; an exceptional hectare is not a substitute for the distribution across 531 million hectares.

A forward evaluation reports stem abundance, basal area, the joint height–architecture distribution, sound material volume, density and carbon fractions, remainder pools, inclusion probabilities and propagated errors. For an independent measured-minus-mapped interval $[R_-, R_+]$ and a prespecified margin $\epsilon \geq 0$, $R_+ < -\epsilon$ supports an overestimated mapped target, $R_- > \epsilon$ supports an underestimated target, and $[R_-, R_+] \subseteq [-\epsilon, \epsilon]$ supports agreement at that precision. An interval meeting none of these criteria is unresolved. Attribution to mapping rather than measurement bias additionally requires the independent reconstruction and its uncertainty to be adequately validated. The interval propagates field, calibration and mapping uncertainty and their dependence; the practical equivalence margin is specified independently before comparison. The archive supplies a measured-record schema and a forward evaluator without populating them with purported class-5 observations. Its available source data support the numerical requirements in Table S1; a class-5 structural survey and its original geometry records have not been recovered for an empirical closure result.

S1.1 Comparisons of biomass and remainder-only scenarios

At an explicitly assumed carbon fraction of 0.47, the printed class-5 mean and the ratio-based value imply approximately 1,130 and 1,140 Mg dry biomass per hectare. Against the cited 395.7 Mg per hectare African network estimate, the ratios are approximately 2.86 and 2.88; the 200–350 Mg per hectare Amazonian range gives different comparison factors [4, 5]. These are contrasts between differently supported estimators, not an independently measured upper bound. Basic density changes the volume requirement, whereas $B = C/f_C$ determines the biomass conversion.

For carbon c_5 , positive material carbon density β and a disjoint remainder $c_R < c_5$, the woody-volume requirement is $V_{\text{req}} = (c_5 - c_R)/\beta$. Lowering β raises this requirement; increasing an independently measured remainder lowers it. The archived 65.3% remainder scenario fixes a hypothetical woody comparator at 395.7 Mg per hectare and assigns the residual elsewhere. It therefore tests remainder-only closure under that assumption, not the primary hypothesis that the woody allometric estimate itself is biased. Foliage measurements in the Higuchi et al. sample and living epiphytes in Gómez González et al. do not support such a large nonwoody fraction in those populations [6, 7]; they do not establish a universal class-5 bound. Woody lianas, small stems and already represented branches are not nonwoody residuals. The source component calculations are retained as numerical diagnostics, without converting this restricted scenario into an explanation imposed on the architectural methodology.

An independently reconstructed plot residual can indicate that either mapped estimator is high, low or compatible within a specified margin. Both paths require validated material inputs and common support; failure of one estimator is not inferred solely from agreement of several products with shared calibration labels. Class-level fields in the companion schema remain unpopulated unless independent records are actually supplied.

S2 Census calibration criteria and derivations

Assume a fixed scalar calibration applies to every event in a matched census. Let j index positive growth intervals $[x_{0j}, x_{1j}]$, $g_j = x_{1j} - x_{0j} > 0$, and d mortality stocks $x_d > 0$. Set $G = \sum_j g_j$, $L = \sum_d x_d$, with weights $v_j = g_j/G$ and $\ell_d = x_d/L$. Recruits may enter with $x_{0j} = 0$ under the specified observed-pool convention. Shrinkage and other losses require additional signed terms, not an unnoticed change in the event set.

Proposition S2.1 (Exact quadratic census criterion). *For $m(x) = \alpha x + kx^2$, with $\alpha > 0$ and $k > 0$ in consistent carbon units, define*

$$\bar{x}_L = \sum_d \ell_d x_d = \frac{\sum_d x_d^2}{L}, \quad \bar{x}_G = \sum_j v_j \frac{x_{0j} + x_{1j}}{2}. \quad (\text{S4})$$

Then

$$r_L - r_G = k(\bar{x}_L - 2\bar{x}_G). \quad (\text{S5})$$

Thus $r_L > r_G$ exactly when $\bar{x}_L > 2\bar{x}_G$; a positive practical margin requires $k(\bar{x}_L - 2\bar{x}_G) > \epsilon_\Delta$.

Proof. For a death, $m(x_d)/x_d = \alpha + kx_d$. For a gain, the exact secant is $\alpha + k(x_{0j} + x_{1j})$. Weighted summation and subtraction prove the result. Neither a small-increment limit nor zero recruitment is assumed. $\square$

The constant scale α cancels from the difference and positive k from its sign, but not from the effect magnitude. The counterexample in Table 7 is exactly $100 < 2 \times 85$. This criterion can screen that calibration family from baseline event data before fitting k ; it is not a calibration-free test of all biological models.

For the power family $m(x) = ax^b$, $a > 0$, $b > 1$ (with the units of a chosen consistently), a finite-interval extension avoids relying on infinitesimal growth. Put $p = b - 1$, and define a probability distribution γ along the gain intervals by

$$\mathbb{E}_\gamma[h(X)] = \frac{1}{G} \sum_j \int_{x_{0j}}^{x_{1j}} h(x) dx, \quad M_{p,L} = (\mathbb{E}_\ell[X^p])^{1/p}, \quad M_{p,G} = (\mathbb{E}_\gamma[X^p])^{1/p}. \quad (\text{S6})$$

Overlapping intervals represent distinct tree events and are counted separately.

Proposition S2.2 (Exact finite-interval power-law criterion). *For these event distributions,*

$$r_L = aM_{p,L}^p, \quad r_G = abM_{p,G}^p, \quad r_L > r_G \iff \frac{M_{p,L}}{M_{p,G}} > b^{1/(b-1)}. \quad (\text{S7})$$

Proof. The loss formula follows by weighting ax_d^{b-1} . The fundamental theorem of calculus gives $x_1^b - x_0^b = b \int_{x_0}^{x_1} x^{b-1} dx$, proving the gain formula. Positive division and the $1/p$ power preserve the inequality. $\square$

The threshold decreases from e as $b \downarrow 1$ to 2 at $b = 2$. Indeed, $\log[b^{1/(b-1)}] = \log(1+p)/p$ decreases for $p > 0$. At $b = 1$, exactly, both factors are a and their difference is zero; the limiting threshold is not a strict-direction test at that endpoint. In the small-increment limit, γ reduces to the event-gain-weighted size distribution. Its p -power mean, not an arithmetic mean substituted at every exponent, remains essential. At $b = 2$, $M_{1,G} = \bar{x}_G$ and the threshold is exactly the quadratic one.

A single arithmetic ratio below 2 does *not* exclude all $1 < b \leq 2$ in heterogeneous censuses. For example, let nine survivors grow from 1 to 1.01, one from 500 to 500.01, and one stock-100 tree die. The ratio $\bar{x}_L/\bar{x}_G = 100/50.905 \approx 1.9644$ fails the quadratic screen. Nevertheless, with $m(x) = x^{3/2}$ the exact factors are $r_L = 10$ and $r_G \approx 4.70749$, so losses are corrected more. This is a synthetic counterexample, not a plausible-sample assertion. The valid screening statistic is exponent-specific: compute the left side of (S7) for the prespecified family. Excluding an entire exponent interval requires the inequality to fail throughout that interval, with numerical tolerances controlled; a few sampled exponents do not prove uniform exclusion. Neither sign screen supplies the calibrated magnitude or independent field support required for H_Δ .

Near-linear calibration. A dimensionally explicit power calibration is $m_b(x) = ax_*(x/x_*)^b$, where $x_* > 0$ is fixed. For $b = 1 + \varepsilon$ and finite logarithmic moments under the loss and gain distributions,

$$r_L - r_G = a\varepsilon [\mathbb{E}_\ell \ln(X/x_*) - \mathbb{E}_\gamma \ln(X/x_*) - 1] + O(\varepsilon^2). \quad (\text{S8})$$

This follows by expanding $(X/x_*)^\varepsilon$ and $(1 + \varepsilon)(X/x_*)^\varepsilon$ in the exact loss and interval-gain operators. Recruitment intervals starting at zero are admissible because $\int_0^h |\ln x|^k dx$ is finite for fixed $h > 0$ and finite k . The differential tends to zero linearly with $b - 1$. Its first-order sign is set by the ratio of loss- and gain-weighted geometric means relative to e ; the reference stock cancels in their log difference. If the coefficient vanishes, higher-order terms determine the sign.

S3 Historical emissions-percentage reconstruction

S3.1 Tracing the 10.3% benchmark

The historical preprint’s emissions discussion cites Ecofys’s *World GHG Emissions Flow Chart 2010*, issued in 2013 [1, 8]. The original chart states a global total of 48,629 MtCO₂e, with all percentages referring to that total. The 10.3% label belongs to *deforestation / afforestation and land-use* CO₂. It is therefore a combined land-use category, not a separately identified gross tropical-deforestation flow.

Its displayed share implies approximately $0.103 \times 48.629 \approx 5.01$ GtCO₂ for that CO₂-labelled component. This is a calculation from rounded graphical inputs, not precision established by independent data. Because the numerator is CO₂, using the multi-gas denominator is dimensionally permissible; the substantive problem is category and source compatibility, not the mere presence of CO₂ inside a CO₂e chart.

The chart cites IEA, EDGAR, CDIAC and Global Carbon Project source vintages. It does not specify a recoverable link from the 10.3% numerator to either the 228.7 or 285.1 PgC stock map, and its plate does not identify a unique GWP assessment basis. The cited plate is the original Ecofys-authored chart preserved by an external mirror. Its original publisher URL and retrieval limitations are recorded in the provenance notes; no unverified Wayback capture is substituted for an accessible source. The exact component assembly and spatial density correspondence therefore remain unverified. These omissions are not filled by assuming that all contemporaneous land-use estimates used the same carbon-density model.

The conversion of a stock contrast into a gross-deforestation percentage requires a common activity definition, spatial basis and stock-to-flux operator. The historical 10.3% category combines land-use effects and removals, so the displayed multiplication is interpreted here as a conditional accounting scenario. These requirements concern the annual inference even if the 723.97 PgC stock is accepted in full. They do not constitute a test that rejects that stock estimate.

S3.2 Two arithmetic reconstructions, not two measurements

Let $s_0 = 0.103$ denote the chart’s combined category share, and provisionally assign it a scalar multiplier r_L . For the arithmetic demonstration only, set $r_L = r_C$ using the displayed stock ratios:

$$r_{C,1} = 723.97/285.1, \quad r_{C,2} = 723.97/228.7. \quad (\text{S9})$$

The fixed-total calculation and the retained-other-contributions calculation are respectively

$$s_{\text{fixed}} = r_L s_0, \quad s_{\text{renorm}} = \frac{r_L s_0}{1 + (r_L - 1)s_0}. \quad (\text{S10})$$

The latter follows by adding the component increment once to the total. The former requires an offset elsewhere to retain that total. Neither establishes that the combined historical land-use component has the imposed multiplier.

Table S2: Reconstruction from displayed historical inputs. The original published range is 26.15–32.60%; nearest rounding of the displayed inputs gives 26.16–32.61%, whereas truncation gives the published 26.15–32.60%. No endpoint is a confidence limit. $r_L = r_C$ is imposed despite unverified spatial and numerator compatibility.

Comparison stock (PgC)	Ratio	Fixed total (%)	Retained others (%)
285.1	2.53935	26.16	22.58
228.7	3.16559	32.61	26.66

The renormalized interval is 22.58–26.66%. The historical fixed-total endpoints are compatible with at least two display paths: truncating the calculation with a baseline of exactly 10.3%, or rounding a calculation with an unprinted baseline share of approximately 10.296665–10.299824%, which itself displays as 10.3%. The percentage-column pattern favors truncation as a coherent explanation, but the source calculation has not been recovered to identify its actual path. Both possibilities are recorded rather than assigning the endpoint difference to an established cause. This distinguishes denominator arithmetic from the much harder stock-to-flux inference. For a harmonized net land component, uniform paired scaling could produce a common multiplier of its calibrated net part, as (25) shows. That mathematical possibility does not establish that Ecofys’s component, the historical map totals and the relevant removal estimates meet those conditions.

Figure 2 therefore puts the two ranges in a labelled historical inset outside the forest ribbon. The green branch identifies forest-carbon remeasurement and paired source–sink revision. The surrounding question marks denote incomplete allocation, not license to transfer an arbitrary amount from energy or any other named sector. The 2010 inset is not a numerical branch of the 2025 inventory.

S4 Published source-table transcription and label diagnostics

Table 3 of the preprint reports 723.97 PgC over 2,799,053,606.91 ha within its mapped pantropical domain, excluding the Australian tropics. Its classes include high-carbon forest, disturbed vegetation, woodland, scrub, savanna and degraded land; the denominator is therefore a mixed vegetation domain, not a rainforest-only area [1]. For a stock C in PgC, area A in hectares and assumed dry-mass carbon fraction f_C ,

$$\bar{c} = 10^9 C / A, \quad \bar{B} = \bar{c} / f_C. \quad (\text{S11})$$

Using the source’s displayed totals gives $\bar{c} = 258.65 \text{ MgC ha}^{-1}$. The two highest-carbon classes contain 566.79 PgC over 1,200,345,923.20 ha, implying $472.19 \text{ MgC ha}^{-1}$. Those carbon densities do not depend on any biomass back-conversion.

The original method reports measured tissue-carbon fractions of 0.4111–0.4999 and predicts carbon directly through its geometric and neural-network calibration [1], rather than specifying a uniform pantropical factor of 0.50. Reconstructing the biomass actually implied by that method requires the associated biomass weights: for compartments j , $B_\Sigma = \sum_j C_j / f_j$ and $f_{\text{eff}} = C_\Sigma / B_\Sigma$. The published aggregate does not recover these weights or an effective pantropical fraction. Neither inversion by a tissue-average fraction nor division by 0.50 reproduces the original biomass in their absence.

As explicit sensitivities, $f_C = 0.47$ gives $550.32 \text{ Mg dry biomass ha}^{-1}$ for the mixed domain and 1004.66 for the high-carbon classes; $f_C = 0.50$ gives 517.30 and 944.38 , respectively. If all mapped compartments additionally had fractions within the measured 0.4111–0.4999 range, the corresponding means would lie within 517.40 – 629.16 and 944.57 – 1148.60 . These are conditional back-conversions, not a recovered biomass surface. In particular, the approximately 1005 Mg ha^{-1} value follows from an assumed 0.47 fraction; it is neither a mandatory output of the original carbon method nor evidence of a physical impossibility. Carbon mass per hectare and the weight of an individual tree are different quantities.

Table S3: Published-digit audit of the historical source’s Table 3 [1]. Intervals, printed means and percentages are transcribed; the ratio column is $10^9 C/A$. Carbon-density units are MgC ha^{-1} . Areas are shown in million hectares here, with the full two-decimal-hectare values in the repository. Classes 2 and 3 fit each other’s intervals. The printed total is 723.97 PgC.

Class	Printed interval	Area (10^6 ha)	PgC	Printed mean	Ratio	Printed %
5	478–605	531.449	285.03	531.05	536.33	39.37
4	336–478	668.897	281.76	417.09	421.23	38.91
2	58–183	419.412	105.54	249.16	251.64	14.57
3	183–336	350.689	40.66	114.80	115.94	5.61
1	1–58	828.607	10.96	13.10	13.23	1.51

The two intermediate numerical rows in the published table are mislabelled relative to its density-class definitions. The row with stock–area ratio $251.64 \text{ MgC ha}^{-1}$ belongs to class 3 (183–336), and the row with ratio 115.94 belongs to class 2 (58–183). Their printed means give the same assignment. Because each numerical pair lies strictly inside one of the existing intervals, direct membership fixes the assignment; in source-row order it is (5, 4, 3, 2, 1) rather than (5, 4, 2, 3, 1). This corrects the printed numerical-row/class association and restores increasing density with class while retaining every stock and area. The source transcription remains separate from the corrected assignment. The operational key is needed to determine whether and where the original vegetation descriptions or raster legend share the labelling problem; the table correction does not establish an error in pixel carbon. Enumeration of the 120 assignments is retained only as a regression check.

The mean-density discrepancy has a common pattern. Define $q_j = \bar{c}_{j,\text{printed}}/(10^9 C_j/A_j)$. The five values in source order are approximately 0.990163, 0.990169, 0.990153, 0.990139 and 0.990397. Under nearest rounding, a value printed to two decimals represents an interval with half-width 0.005; under positive-number truncation, it represents the interval from the printed value to that value plus 0.01. Applying the respective interval rules to stock, area and mean gives common-factor bounds [0.990140, 0.990190] and [0.990134, 0.990182]. Both models therefore accommodate one shared normalization. Exact division by 1.01 is excluded by classes 4 and 5 under either rule; the available digits identify an approximately one-percent convention difference, rather than its computational origin.

All five numerical rows are compatible, within displayed precision, with one common multiplicative normalization; separate class-specific factors are unnecessary. The representative $q = 0.99016$ gives $1/q = 1.0099378$. Under a shared carbon numerator, $q = A_{\text{table}}/A_{\text{mean}}$, so a denominator about 0.994% larger would reproduce the central factor. A common multiplier is a parsimonious description, but its processing stage is not identified. In particular, a spatial area multiplier $s(x) > 0$ gives $q_j = A_j/\int_{\Omega_j} s(x) dA = 1/\mathbb{E}_{A,j}[s]$. Different class footprints can have the same class-weighted mean even when s varies within them. Thus a spatial explanation must reproduce the common class-level factor, rather than being categorically excluded. Class 1’s central-factor residual is only $0.00314 \text{ MgC ha}^{-1}$, below half a displayed mean-density unit. Its weaker relative precision supplies no separate proof of a projection effect. Mean-column operations, pixel-area weights and denominator records distinguish the alternatives.

There is direct evidence for truncation in the percentage column. Truncating $100C_j/723.97$ to two decimals reproduces all five percentages in Table S3; nearest rounding does not reproduce classes 4, 2 and 3. The stock sum is 723.95 PgC against the printed 723.97, and the area sum is 2,799,053,606.89 ha against the printed 2,799,053,606.91. Both discrepancies are two units of the last displayed digit. These results are consistent with truncating component values and a separately computed total.

A probability calculation can compare display conventions only after its assumptions are specified. For

five independent uniform hidden component fractions and a total computed from their exact sum under the same display rule, a positive two-unit gap has probability $11/20 = 0.55$ under truncation and $1/120$ under nearest rounding; the probability of a gap of either sign with magnitude two under nearest rounding is $1/60$. These are conditional reference-model probabilities, not posterior probabilities of the historical workflow. Carbon and area outputs may share computations, so their probabilities are not multiplied. Truncation explains the visible pattern parsimoniously, while the original calculation is needed to identify the actual display convention.

The printed-table correction, common-factor compatibility and historical processing origin are separate results. The first is determined by interval membership and leaves carbon quantities unchanged. The second follows from the digit intervals. The third requires the original workbook, class key and area computations, which were not recovered among the available records. The source transcription and corrected class association are supplied separately in the numerical archive.

Spatial support determines the meaning of a comparison. Feldpausch et al.'s 285.1 PgC total uses 1667.5 million hectares of specified forest classes, with a mean of $170.97 \text{ MgC ha}^{-1}$ [2]. For any two stock–area totals $C = A\bar{c}$, their ratio decomposes as

$$\frac{C_A}{C_B} = \frac{A_A}{A_B} \frac{\bar{c}_A}{\bar{c}_B}. \quad (\text{S12})$$

The complete historical comparison is approximately $2.539 \approx 1.679 \times 1.513$: both the larger area and the different mean density contribute to the numerical ratio. This is an arithmetic factorization, not a causal attribution of the difference. The broader historical domain includes extensive lower-carbon vegetation, so its mixed-domain mean attenuates the high-class density contrast.

Classes 4 and 5 contain 566.79 PgC on 1200.346 million hectares: 1.988 times Feldpausch's entire stock on 0.720 times its area. Their mean density is 2.762 times the whole-domain comparison mean. This highlights the high-density requirement but is not yet a matched-pixel correction factor. Let H denote those classes and $C_{0,H}$ their conventional carbon. If H is contained in Feldpausch's mask, then $0 < C_{0,H} \leq 285.1$ and

$$r_H = \frac{566.79}{C_{0,H}} \geq \frac{566.79}{285.1} \approx 1.99. \quad (\text{S13})$$

The value 2.76 is obtained if H had the conventional whole-domain mean density. If its conventional mean is additionally at least that whole-domain mean, the conditional bracket becomes $1.99 \lesssim r_H \lesssim 2.76$. Without this extra density assumption, 2.76 is a comparison scenario, not an upper bound: a lower conventional local density can require a larger factor. Neither nesting nor that density ordering has been established by raster intersection here. The repository supplies explicit counterexamples to an unconditional finite bracket.

A common-mask analysis is needed to estimate $C_{0,H}$ and separate biological calibration from vegetation selection. The comparison with 228.7 PgC also requires matched pools, vegetation support and period [9]; no associated Baccini area has been recovered for this audit. Equation (S12) is therefore evaluated for Feldpausch only, while the Baccini historical endpoint remains a total-stock arithmetic reconstruction.

S5 Compact statistical and methodological notes

S5.1 Reconciliation and structural identifiability

For an observation model $y = Hx + \epsilon$ with parameter covariance or likelihood specified independently, a linear combination $a^T x$ is identifiable from noiseless Hx exactly when a belongs to the row space of H . To see this, two parameter vectors with the same observation differ by $v \in \ker H$. Their target values agree precisely when $a^T v = 0$ for every such v , equivalently $a \in (\ker H)^\perp$. This is the general form of the source–sink ambiguity in (31).

Under an independently specified Gaussian reference x_0 with covariance Σ and an observation-error covariance V , the unconstrained conditional mean is

$$\hat{x} = x_0 + \Sigma H^T (H \Sigma H^T + V)^{-1} (y - H x_0). \quad (\text{S14})$$

This standard result does not add information absent from the covariance and observations. Hard physical bounds or non-Gaussian discrepancies require an appropriate constrained or non-linear analysis. A statistically unique posterior mean produced by a prior is not the same as structural identification by the data. Shared allometric calibration belongs in covariance or explicit model-discrepancy terms, not in a repeated claim of independent evidence.

S5.2 Bias, covariance and ratios

For a squared-error estimator, $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$. Additional samples can reduce sampling variance without eliminating a common model error. Inventory uncertainty may involve activity data, emission factors and correlated aggregation [10]. The paired correction is a structured instance of such dependence, not a claim that all uncertainties are equal or symmetric.

For a ratio $s = F/T$ with nonzero T , first-order propagation gives

$$\text{Var}(s) \simeq \frac{\text{Var}(F)}{T^2} + \frac{F^2 \text{Var}(T)}{T^4} - \frac{2F \text{Cov}(F, T)}{T^3}. \quad (\text{S15})$$

If T contains F , independence is generally unjustified. The approximation is unreliable near a zero or poorly constrained denominator. Joint simulation or explicit ratio inference is then preferable. The present article does not fit an uncertainty interval around the historical reconstructed percentages.

S5.3 Model domains and admissibility

A unit with zero baseline carbon and zero corrected contribution may be omitted from the restricted fixed-stock release representation. That permission does not extend to regrowth from zero stock, positive corrected flux on a zero baseline increment, or newly represented carbon pools. Those cases require the increment or additive terms in Section 7.2. Newly included areas are a support change and must be accounted for separately. Positive baseline flux is necessary to define a multiplier. Signed net weights do not satisfy the convex-combination bounds for r_F ; separating positive source and removal components avoids that misuse. The general net correction in (19) does not require common calibration; only its common- r_C expansion does. Separate bases admit two transfer identities, but no common covariance term follows without an explicit mapping. The algebraic identity for a net correction remains valid when its source and removal ratios are supplied by different admissible operators; the common-stock covariance formula does not.

For repeated censuses, additions and removals of sampling units must be reconciled before invoking scalar change identities. For delayed release, the cohort index can absorb spatial position, disturbance year, tree-size compartment, organ and carbon pool. A change in decomposition kernel, combustion completeness or regrowth trajectory is a process correction, not merely a carbon-stock multiplier. These distinctions are necessary both for causal interpretation and for conservation of carbon in an implementation.

S6 Published-data application and residual identifiability

The input audit uses the *Acalypha* sp.–*Guazuma ulmifolia* community in Vásquez and Arellano [11, Table 170, p. 942]. The published observations are community summaries per 0.05 ha, rather than a new single-plot survey. The source-method carbon combines allometric biomass and measured tissue fractions. Those quantities are retained under their actual measurement status.

Table S4: Worked audit of real published Córdoba aggregates. Per-hectare results divide source totals by 0.05 ha. Class height ranges bound the basal-area–height cylinder sum from displayed inputs; the sum is not solid woody volume. Minor aggregate differences follow displayed precision.

Quantity	Reproduced result
Reported mean stem density	1,593.4 ha ⁻¹
Reported mean basal area	45.2 m ² ha ⁻¹
Sum of displayed class basal areas	45.4 m ² ha ⁻¹
Allometry-based aboveground biomass	147.8 Mg ha ⁻¹
Source-method carbon	66.8 MgC ha ⁻¹
Displayed-class $\sum_h G_h H_h$ envelope	236.5–778.4 m ³ ha ⁻¹
Independent geometric carbon / mapped residual	Not identifiable

With reported class basal areas G_h and height limits $H_{h,-}, H_{h,+}$, the diagnostic is

$$\sum_h G_h H_{h,-} \leq \sum_j a_j H_j / A \leq \sum_h G_h H_{h,+}.$$

Applied to displayed class aggregates it gives a nominal cylinder-sum envelope, not an error interval or a woody-volume estimate. It does not supply the unknown joint distribution of ψ , density and chemistry. The sum of the rounded class basal areas is 45.4 m² ha⁻¹, versus the separately printed total 45.2; neither is silently substituted for the other.

The source summary lacks independent sound organ volumes, their paired material properties, and a recovered map value and class attached to the same georeferenced footprint and epoch. Converting its allometric biomass back into volume would recycle the estimator under test; assigning a mapped class from the source-method carbon would assume the desired map correspondence. Accordingly, `data/cordoba_empirical_audit.json` records the reproduced quantities, absent inputs and a null geometric residual. The executable eligibility check thus prevents an unsupported empirical claim. The separately reported water-displacement comparison tests reconstructed volume at its sampled scale and does not furnish the missing class-level carbon residual.

S7 Evidence from harvest, laser scanning and forest networks

Reference methods require the same scrutiny as the allometries under test. Gonzalez de Tanago et al. [12] used nine of 29 harvested trees for reconstruction calibration and 20 for independent validation. Their Equation 3 defines bias by the mean residual divided by mean reference AGB, equivalently

$$b = \frac{\sum_j (\widehat{B}_j - B_j^{\text{ref}})}{\sum_j B_j^{\text{ref}}}, \quad \frac{\sum_j B_j^{\text{ref}}}{\sum_j \widehat{B}_j} = \frac{1}{1 + b}. \quad (\text{S16})$$

Thus $1/(1 - d)$ is valid for a reported negative bias $b = -d$, but as a ratio of *sample totals*, not the mean of individual replacement factors. Their Table 4 explicitly identifies 20 validation trees and gives endpoint factors of approximately 1.180–1.554. Sampling selected the largest tree in each of 29 plots (mean diameter 73.5 cm across the sampled trees). The reported increase of allometric underestimation with size cautions against extrapolating this large-tree sample to a stand: a smaller-tree-weighted population would have a lower aggregate factor only if the same size-dependent calibration and relevant weighting also apply. The reference AGB itself was calculated from harvested-tree geometric measurements, assigned wood density and small-branch expansion, not by weighing all reference biomass; the authors explicitly acknowledge those limitations. This is informative structural validation, but not independent validation of every density or chemical input. Momo Takoudjou et al. [13] likewise demonstrate the importance of checking reconstruction quality rather than declaring laser-derived models error-free.

Table S5: Selected primary evidence and the limit of its transfer to a global correction. Percentages have different denominators and must not be pooled as interchangeable effect sizes. Dry-mass densities are in Mg ha^{-1} .

Study / support	Reported result	Interpretation limit
Gonzalez de Tanago et al. (2018); 29 trees, of which 20 formed the independent validation set	Harvest-geometry reference: TLS bias -3.68% ; pantropical-model biases -15.22 to -35.67% .	Geometry and assigned density, not complete mass weighing. Aggregate validation-sample replacement factors approximately 1.180–1.554.
Momo Takoudjou et al. (2018); 61 harvested trees, Cameroon	Edited quantitative structure models had 4.68% volume bias; TLS- and harvest-calibrated allometric coefficients were statistically indistinguishable.	The quoted percentage is <i>volume</i> , not global carbon bias. Manual model editing and density remain consequential.
Huertas et al. (2025); 118.75 ha of French Guianan plots	Local species-specific ALS height–diameter models produced 11–13% higher AGB estimates.	Local model comparison; not independent weighing of a pantropical carbon stock.
Demol et al. (2024); miombo landscape	Carbon estimates 1.5–2.2 times conventional comparison estimates at landscape scale.	Dry woodland and scaling methods differ from moist-forest and whole-tropics inference.
Kauffman et al. (1995); four slashed Amazonian primary forests	Total aboveground fuel biomass 292–436; forest-floor material is included.	Clearing-site inventories are not equivalent to complete, independent weighing of all live trees in one hectare. Pools differ from live AGB.
Malhi et al. (2006); Amazonian plot-network synthesis	Regional aboveground live biomass approximately 200–350, with spatial variation and additional small-stem/liana accounting.	Allometry-dependent spatial estimates, useful as contextual comparison, not an independent absolute ceiling.
Lewis et al. (2013); 260 intact African plots	Mean AGB 395.7; 95% confidence interval on the mean ± 14.3 .	Intact closed-canopy sample, not all vegetation; reference calibration can be shared.
Feldpausch et al. (2012); height-integrated tropical analysis	Including height reduced the estimated total from 320.5 to 285.1 PgC over the same forest mask.	A counter-direction model result; compared masks must remain identical.
Avitabile et al. (2016); fused pantropical map	Fusion and bias adjustment gave a lower total than the input-map estimates.	Independent map differences can involve support, calibration and fusion, not one universal biological multiplier.

The moist-forest local-calibration and miombo findings demonstrate that substantive differences can arise from architecture, species calibration and mapping scale [14, 15]. The mixed-domain density contrast of approximately 1.51 lies numerically within the Gonzalez de Tanago and miombo ranges. Under nested masks, the high-class comparison gives a lower factor near 1.99; the 2.76 scenario additionally assigns a whole-domain density to that subset. These comparisons describe different populations, supports and estimators. A regional factor is neither confirmation of a pantropical total nor an upper bound on the correction permitted elsewhere. Height integration and map fusion can also move estimates downward [2, 16], emphasizing that each result must be interpreted through its calibration chain.

The network contrasts are quantified in Supplementary Note S1: class-5 conditional biomass is 2.88 times the cited African mean and 3.26–5.71 times the Amazonian regional range [4, 5]. Their allometric dependence and different supports require an independent class-level material comparison, not omission of the gap. Kauffman et al.’s 292–436 Mg ha⁻¹ clearing-site range refers to aboveground fuel pools [17]; it must be distinguished from the live-biomass network quantities. The forward protocol tests the relevant carbon-volume relation with independent geometry, chemistry, compartment completeness and spatial selection.

A genuine whole-hectare destructive mass census would strengthen this comparison, but a hectare-scale clearing experiment must not be relabelled as such solely because its footprint is one hectare. Inventory-derived initial masses and sampled burn consumption remain model-assisted estimates. Here harvested-tree geometry comparisons, any directly weighed reference material, clearing inventories and plot networks remain distinct evidential classes. This distinction prevents an apparently independent benchmark from covertly reusing the calibration being tested.

S8 Reproducibility and source provenance

S8.1 Inputs, outputs and numerical checks

The companion archive contains the exact displayed source entries in `data/inputs.json`, including class order, full-precision areas, means, percentages and intervals. It also records the Ecofys category and denominator, the Gonzalez de Tanago validation metric and explicitly synthetic examples. The published table is left unchanged by the diagnostic calculations.

`scripts/reproduce.py`, `scripts/extensions.py` and `scripts/audit_extensions.py` regenerate the numerical and LaTeX-table outputs listed in the manifest. These cover density and support comparisons, crossed intervals, nearest-rounding and truncation models, exact conditional gap probabilities, paired accounts, zero-stock regrowth, finite-increment census criteria, single- and mixed-cohort release, near-linear calibration and invalid inputs. Machine-readable values and typeset tables share the same calculations.

The core suite comprises 1,000 baseline cases (seed 20260916), 20,000 extension cases (seed 20260917) and 1,100 audit-extension cases (seed 20260918), with 257,167 assertions across `reproduce.py`, `extensions.py` and `audit_extensions.py`. That count excludes the four later modules: `population_atmosphere.py` (40 deterministic checks), `physical_closure.py` (1,000 synthetic-plot cases, seed 20260919; 6,013 assertions), `class5_comparison.py` (55 deterministic checks) and `admissibility.py` (250 synthetic cases, seed 20260921; 515 assertions). Together these five reported suites contain 263,790 assertions/checks; `measurement_evidence.py` additionally performs one regression assertion on the physical-validation shortfalls. Algebraic comparisons use absolute tolerance 10^{-10} and relative tolerance 10^{-12} where applicable; decimal intervals and conditional probabilities use high-precision decimal or exact rational arithmetic. The near-linear expansion is tested through its scaled second-order remainder. Each validation record identifies its own count, scope and residual.

The additional `scripts/population_atmosphere.py` regenerates the intermediate-stem table,

original near-100 kg records and paired atmospheric-account examples from `data/population_atmosphere_inputs.json`. Its separate validation report checks mass units, aggregate means, abundance-weighted contributions, delayed release and source-ranking conditions. These deterministic checks reproduce displayed inputs and algebra, rather than identifying a pantropical mass distribution or fitting an atmospheric trajectory.

The module `scripts/physical_closure.py` adds forward plot-level material reconstruction, the class-5 conditional requirements, a regression enumeration of the direct interval assignment, an area-denominator diagnostic and the community-density conversion. Its separate checks use seed 20260919 and test the closure identity, area scaling, population weighting and input rejection; they are computational tests on synthetic records. `data/stand_measurement_schema.json` specifies the measured input format and `scripts/physical_closure.py--measurements` evaluates supplied records without fitting them to the mapped target. A separate output file is required for such an evaluation, keeping measured results distinct from the illustrative requirements.

The additional `scripts/class5_comparison.py` computes both density-convention requirements and their differences from the source-reported African and Amazonian values. It checks the directions of the volume sensitivities, the independent-pool mass constraint, direct interval membership and a spatial-area counterexample. Inputs have source or illustrative-status fields; no new field record is generated. Its separate deterministic validation record supplements, rather than replaces, the earlier analytical tests.

The module `scripts/admissibility.py` generates the lower-edge and dual-mean material requirements, the weighted height–architecture decomposition, component-evidence ratios, the stated remainder screen and the Córdoba input audit. It also tests both signs of an independent measured-minus-mapped interval. Public conditional labels are rounded; exact computational values and source transcriptions remain in machine-readable files. Its synthetic tests establish algebraic and input-check consistency, not an observed mapped-target correction.

S8.2 Archive and repository verification

The repository manifest lists relative paths, byte counts and SHA-256 digests for the research snapshot. The verifier checks integrity and compares regenerated numerical outputs within stated cross-platform floating-point tolerances; non-numerical values and CSV/LaTeX text remain exact comparisons. Build reports identify compiler versions, warnings and PDF preflight results. These are integrity and computational-reproduction checks, rather than digital-authenticity certification or biological validation. Source-observation years, publication years and the research-snapshot date are recorded separately.

S8.3 Empirical scope and unrecovered records

The numerical archive reconstructs published tables and tests exact identities. It includes a source-transcribed intermediate-stem example, near-100 kg architectural records and deterministic atmospheric-account demonstrations; source-reported shares remain distinct from derived means. A common-mask carbon integration, original classification-pipeline rerun, reconstructed Ecofys component inventory and matched field census analysis require additional empirical records. The source-availability note identifies the workbook, raster, class-key and carbon-fraction weights that were not recovered, without assigning an unverified reason for their absence. The table-level interval assignment is unique under the stated fixed scheme. Its translation to a historical raster legend, and the computational cause of the common normalization, require the original records.

S8.4 Calibration-label and population-transfer demonstrations

The additional script `scripts/calibration_bias.py` evaluates the conditional label-inheritance, population-transfer and shared-offset identities in finite synthetic populations. It includes deterministic counterexamples, 1,000 seeded randomized cases and independent recomputation of both sides. It uses seed 20260922 and records 8,014 assertions, in addition to the prior suites. All 54 numerical/table outputs are regenerated by the author-package verifier; the publication-only verifier excludes LaTeX fragments from its distributed files and rebuilds them only in a temporary working directory. Its outputs separate signed bias from prediction variance and include cases of zero and nonzero inherited conditional bias. These calculations are analytical demonstrations, not measurements of the bias of GEDI or the published pantropical map.

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